ETHICAL AI · CONSOLIDATED SUMMARY NOTES

Unit 2 — Bias and Fairness

Bias in data & models • Impact on decision making • Types of bias • Real-world cases • Techniques to detect & mitigate bias • Implementing bias detection and fairness

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Source: **Introduction to Responsible AI** (Manure, Bengani, Saravanan, Apress 2023) — Ch. 2 *Bias and Fairness*. Figures 2-1 to 2-4 are from the book.

1. Understanding Bias in Data and Models

Bias in data and models

The presence of **systematic deviations** that lead to inaccuracies or unfairness in decision-making. It emerges when data-collection or model-construction processes **inadvertently favour certain groups, attributes or perspectives** over others.

- **Sources:** historical inequalities, flawed data-collection methods, biased algorithms.
- **Addressing it needs:** a deep understanding of how bias shows up in **both data and model outcomes**, plus strategies that ensure equitable, unbiased decision making.
- Detecting and mitigating bias is empowered by technology; the goal is transparent, responsible AI that champions diversity and societal progress.

2. Importance of Understanding Bias

Why	Explanation
Avoiding discrimination & inequity	Bias in data or models can produce discriminatory outcomes; understanding its origins lets developers treat all individuals impartially regardless of background, gender or race.
Ensuring ethical AI deployment	Lets developers align work with ethical principles and legal mandates — steer clear of accentuating societal biases; strive for fairness, transparency, accountability.
Building trust	Systems that keep producing biased outcomes erode public trust; addressing bias builds credibility and the belief that AI functions equitably.
Better decision making	AI now drives hiring, lending, criminal justice; understanding bias keeps those decisions informed, transparent and free of discriminatory influence.
Promoting innovation	A system tainted by bias may fail to represent the spectrum of human experience; fixing bias makes AI adaptive, versatile and potent across contexts.

Reducing reproduction of historical injustices	AI learning from biased data can inherit negative historical patterns and stereotypes; understanding this enables proactive prevention.
Diversity & inclusion	Pushes teams to be a microcosm of diversity → more comprehensive system design.
Social progress & long-term viability	Fair AI bridges disparities and champions equal opportunity; equitable systems enable sustainable adoption in health care, education and finance.

3. How Bias Can Impact Decision-Making Processes

- **Distorted perceptions** — bias changes how information is perceived; some aspects over-weighted, others missed → skewed understanding.
- **Unconscious (implicit) biases** — cultural, societal and personal factors shape judgments even in well-intentioned people.
- **Confirmation bias** — favouring information that confirms existing beliefs; contradictory information is ignored.
- **Stereotyping** — assumptions about a person based on group membership instead of individual merit.
- **Unequal treatment** — different opportunities, resources or punishments by race, gender, socioeconomic status.
- **Discriminatory outcomes** — at individual and systemic levels: access to education, employment, health care.
- **Impact on automated systems** — biased training data → biased predictions; entrenches inequality.
- **Feedback loops** — biased decisions limit a group's opportunities, reinforcing stereotypes and further marginalising them.
- **Erosion of trust** — perceived bias damages trust in institutions, can cause social unrest.
- **Reinforcing inequalities** — groups that keep facing biased decisions stay in a cycle of disadvantage.

4. Types of Bias

Bias in ML = systematic, unfair errors in data or models that lead to inaccurate or unjust predictions. It can appear at different stages of the ML pipeline (Fig. 2-1).

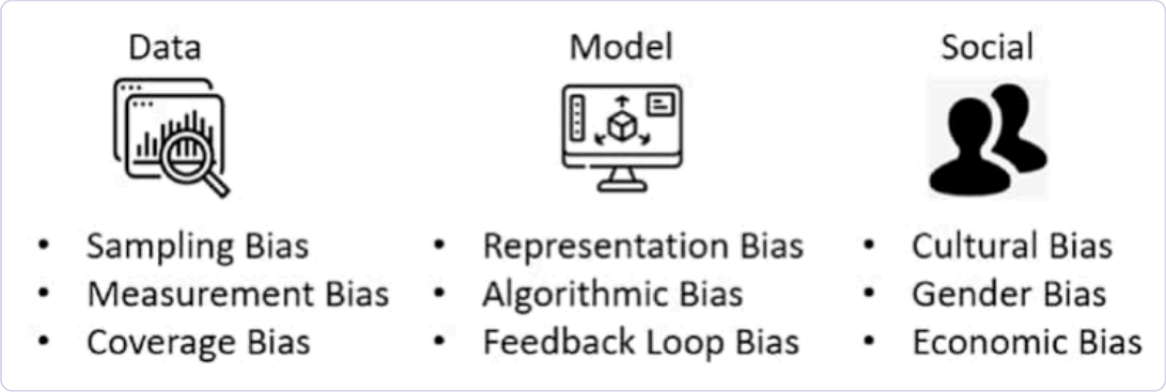


Figure 2-1 — Types of bias: Data, Model and Social

Category	Type	Meaning / example
1. Data bias	**Sampling bias**	Collected data isn't representative of the population → over/under-representation. E.g. a medical dataset with only one demographic group → poor performance on others.
	Measurement bias	Errors or inconsistencies in data collection/measurement. E.g. a survey in a language a community doesn't understand omits their view.
	Coverage bias	Certain groups or perspectives are missing (biased collection methods, incomplete sampling, systemic exclusion).

2. Model bias (from reliance on biased data in training; may be amplified)	**Representation bias**	Training features/attributes disproportionately favour certain groups; the model learns those biases.
	Algorithmic bias	Some algorithms inherently perpetuate bias, e.g. a decision tree splitting on biased features reflects them in predictions.
	Feedback-loop bias	Predictions influence real-world decisions that shape future training data → bias reinforced over time.
3. Social bias (society's biases reflected in data and models)	**Cultural bias**	Cultural norms, beliefs and values shape how data is collected and interpreted.
	Gender bias	Historical gender roles → unequal representation in datasets, affecting performance.
	Racial bias	Biased historical practices → under- or misrepresentation of racial groups.
	Economic bias	Socioeconomic disparities → differences in data availability and quality.

5. Examples of Real-World Cases Where Models Exhibited Bias

Case	What happened
1. Amazon's gender-biased hiring tool (2018)	AI recruiting tool trained on ten years of resumes, mostly from men → learned to favour male applicants and **downgrade resumes containing terms associated with women**.
2. Racial bias in criminal risk assessment	Tools predicting reoffending from historical arrest and conviction data **overestimate risk for minority groups**, leading to discriminatory sentencing and parole decisions.
3. Google Photos racist labelling (2015)	Auto-tagging labelled images of Black people as **"gorillas"** — training data lacked diverse examples; shows the need for inclusive datasets.
4. Biased loan-approval models	Some models unfairly denied loans to minority applicants or offered higher interest rates, **reflecting historical bias in lending data**.
5. Facial recognition & racial bias	More likely to **misidentify people with darker skin tones, particularly women** → inaccurate surveillance, racial profiling, infringement of civil rights.

Takeaway
Curate diverse, representative training data, use fairness-aware algorithms, apply bias detection & mitigation, and continuously monitor model outputs for fairness.

6. Techniques to Detect Bias

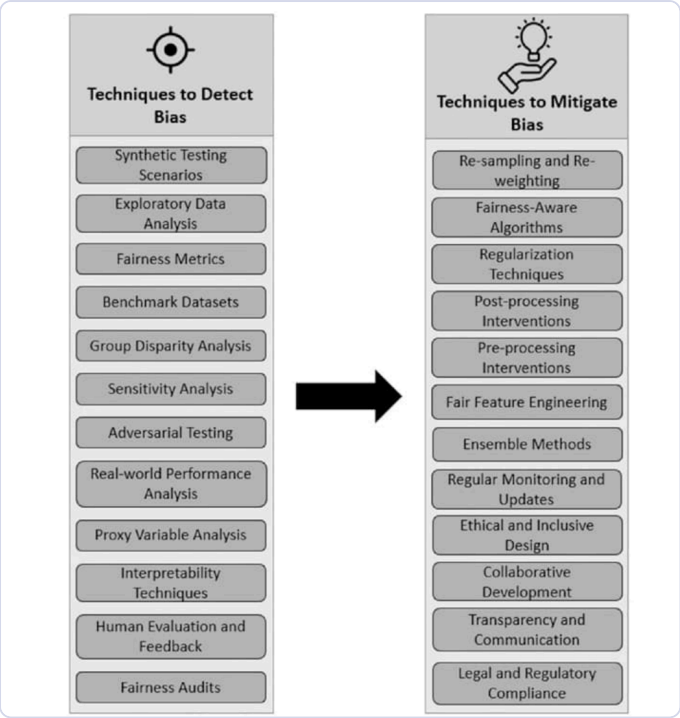


Figure 2-2 — Techniques to detect and mitigate bias

Technique	How it detects bias
Exploratory data analysis (EDA)	Visualise distributions and patterns across groups/attributes to spot disparities.
Fairness metrics	Quantify bias in predictions — **disparate impact**, **equal opportunity difference**, **statistical parity difference** — comparing outcomes between groups.
Benchmark datasets	Datasets with intentionally present fairness issues, to test how well models handle bias.
Group disparity analysis	Compare outcomes (acceptance rates, loan approvals, hiring) across groups and attributes.
Sensitivity analysis	Test how small changes in inputs affect outcomes; reveals features driving biased predictions.
Adversarial testing	Deliberately feed biased inputs to see how the model responds and how susceptible it is.
Real-world performance analysis	Monitor deployed models continuously to catch emerging bias patterns.
Proxy variable analysis	Check attributes correlated with protected characteristics (gender, race) — they can hide bias.
Interpretability techniques	E.g. feature-importance analysis shows which features drive predictions; disproportionate biased features signal bias.
Human evaluation & feedback	Diverse reviewers inspect outputs and spot bias automated checks miss.
Fairness audits	Comprehensive review of data collection, preprocessing and model development.
Synthetic testing scenarios	Controlled synthetic data simulates potential bias sources to observe their impact.

Key fairness metrics

Statistical parity difference = $P(\hat{Y}=1 \mid \text{unprivileged}) - P(\hat{Y}=1 \mid \text{privileged})$ (ideal 0) • **Disparate impact** = $P(\hat{Y}=1 \mid \text{unprivileged}) \div P(\hat{Y}=1 \mid \text{privileged})$ (ideal 1; < 0.8 is flagged) • **Equal opportunity difference** = $TPR_{\text{unprivileged}} - TPR_{\text{privileged}}$ (ideal 0) • **Average odds difference** = mean of the FPR and TPR differences between groups (ideal 0).

7. Techniques to Mitigate Bias

#	Strategy	Detail
1	**Re-sampling**	Balance class representation: **oversample** under-represented or **undersample** over-represented groups.
2	**Re-weighting**	Assign different weights to classes/samples to address imbalance in learning.
3	**Fairness-aware algorithms**	**Adversarial debiasing** (an adversarial network forces the main model to disregard bias-correlated features); **equalised odds** (adjust thresholds for equal opportunity across groups); **reject-option classification** (decline to predict when fairness is uncertain).
4	**Regularisation techniques**	**Fairness constraints** in the optimisation; **Lagrangian relaxation** to balance fairness vs accuracy via Lagrange multipliers.
5	**Post-processing interventions**	**Calibration** of predictions to fairness criteria; **re-ranking** predictions to promote fairness without hurting accuracy.
6	**Pre-processing interventions**	**Data augmentation** (synthesised data for under-represented groups); **de-biasing preprocessing** (re-weighting / re-sampling before training).
7	**Fair feature engineering**	Create/select features less correlated with bias.
8	**Ensemble methods**	Combine models trained with different strategies — biases in individual models are less likely to coincide.
9	**Regular monitoring & updates**	Keep checking real-world performance and update the model as new data arrives.
10	**Ethical & inclusive design**	Diverse representation and ethical considerations in data collection, preprocessing and model development.
11	**Collaborative development**	Involve stakeholders from diverse backgrounds, ethicists and affected communities.
12	**Transparency & communication**	Be open with users and stakeholders about mitigation steps to build trust.
13	**Legal & regulatory compliance**	Adhere to anti-discrimination and bias laws.

Pre-processing (data) → In-processing (algorithm / constraints) → Post-processing (outputs)

8. Implementing Bias Detection & Fairness — Stage 1: Data Bias

Goal: train on the original data, detect racial bias by analysing it, then **resample** for a fair representation and retrain. **Dataset:** UCI **Adult** census income data (48,842 rows, 14 features + target `income` >50k / ≤50k) — age, workclass, education, marital-status, occupation, relationship, **race**, **sex**, capital-gain/loss, hours-per-week, native-country.

Import → Load CSV → Check (info) → Encode → Plot distributions → RandomForest → Resample → Retrain

1. **Packages:** pandas, numpy, `train_test_split`, `RandomForestClassifier`, accuracy/classification_report, `sklearn.utils.resample`, `LabelEncoder`.
2. **Load & check:** `df = pd.read_csv("Income.csv")`, `df.info()` → no null values.
3. **Preprocess: label-encode** categorical columns; **one-hot** (`pd.get_dummies`) columns with more than two categories.
4. **Explore:** value-count bar plots show the imbalance (below).

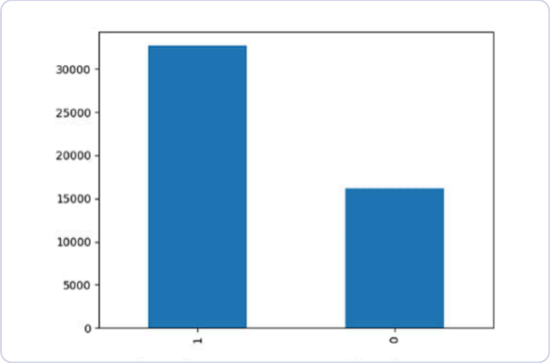


Figure 2-3 — Gender distribution: ~67% male vs 33% female

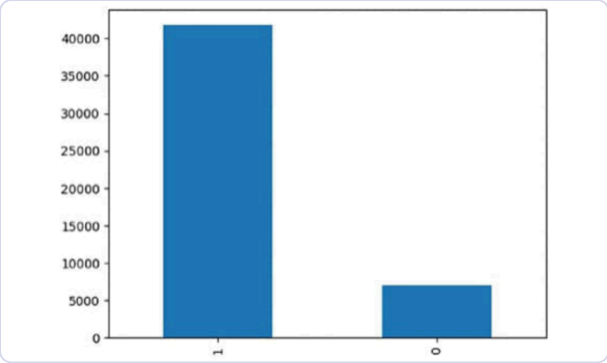


Figure 2-4 — Race distribution: ~83% white vs 17% Black

```
x = df.drop(['income'], axis=1);  y = df['income']
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=42)
model = RandomForestClassifier(random_state=42).fit(x_train, y_train)
print(classification_report(model.predict(x_test), y_test))   # accuracy 0.8526

# Mitigate: upsample the minority race group (race == 0, Black) to match the majority
black = df[df['race'] == 0];  white = df[df['race'] == 1]
upscaled_black = resample(black, replace=True, n_samples=len(white), random_state=42)
balanced_data = pd.concat([upscaled_black, white])
# ...split balanced_data and retrain the same RandomForest
```

Model	Accuracy	F1 (class 0 / class 1)	Observation
Original (biased) data	85.3%	0.91 / **0.67**	Much weaker on the minority/positive class due to imbalance
After resampling	97.4%	0.98 / **0.93**	Class-1 precision 0.92 and recall 0.95 — large improvement

> **Exam-smart caution (not in the book)**
> The 97% is inflated: upsampling *with replacement* before the train/test split puts duplicate rows in both sets (data leakage). In practice, split first and resample **only the training set**.

9. Implementing — Stage 2: Model Bias (AI Fairness 360)

Goal: detect and mitigate bias *in the model* using IBM **AIF360**'s **Adversarial Debiasing** — trains **two neural networks adversarially**: a classifier predicts the label while an adversary tries to predict the **protected attribute** from the output; the classifier is penalised if it can, which removes dependence on the protected attribute. Works for binary classification with a binary protected attribute (gender, race, age). It is promising but only one solution.

Import AIF360 → Load preprocessed Adult → Train plain model (debias=False) → Train debiased model (debias=True) → Compare metrics

```
from aif360.datasets import AdultDataset
from aif360.metrics import ClassificationMetric
from aif360.algorithms.inprocessing.adversarial_debiasing import AdversarialDebiasing
import tensorflow.compat.v1 as tf;  tf.disable_eager_execution()

df = load_preproc_data_adult()
privileged   = [{'sex': 1}];  unprivileged = [{'sex': 0}]
df_train, df_test = df.split([0.7], shuffle=True)

session = tf.Session()
plain = AdversarialDebiasing(privileged_groups=privileged, unprivileged_groups=unprivileged,
                             scope_name='plain_classifier', debias=False, sess=session)
plain.fit(df_train)                        # ... then close session, reset graph

session = tf.Session()
debiased = AdversarialDebiasing(privileged_groups=privileged, unprivileged_groups=unprivileged,
                                scope_name='debiased_classifier', debias=True, sess=session)
debiased.fit(df_train)
m = ClassificationMetric(df_test, debiased.predict(df_test),
        unprivileged_groups=unprivileged, privileged_groups=privileged)
m.accuracy(); m.disparate_impact(); m.equal_opportunity_difference(); m.average_odds_difference()
```

Metric (book's output)	Without debiasing	With debiasing	Ideal
Accuracy	0.809	0.796	—
Disparate impact	0.000	**0.562**	1
Equal opportunity difference	−0.461	**−0.079**	0
Average odds difference	−0.283	**−0.047**	0

Interpretation
Debiasing sharply reduces the gap in true-positive rates and in true/false-positive rates between protected and unprotected groups, at only a **small accuracy cost** (0.809 → 0.796) — the **fairness–accuracy trade-off**. The book notes the metrics improve though accuracy slightly drops.

10. Quick Revision & Likely Questions

One-liners

- **Bias** = systematic deviation causing unfair outcomes; from data collection, model construction, history, society.
- **Types: Data** (sampling, measurement, coverage) • **Model** (representation, algorithmic, feedback loop) • **Social** (cultural, gender, racial, economic).
- **Impact:** distorted perception, stereotyping, unequal treatment, feedback loops, erosion of trust, reinforced inequality.
- **Cases:** Amazon hiring • criminal risk assessment • Google Photos • loan approval • facial recognition.
- **Detect:** EDA, fairness metrics (disparate impact, equal opportunity diff., statistical parity diff.), group disparity, sensitivity, adversarial testing, proxy analysis, interpretability, audits.
- **Mitigate:** re-sampling / re-weighting (pre), adversarial debiasing / equalised odds / constraints (in), calibration / re-ranking / reject option (post) + diverse teams, monitoring, compliance.
- **Implementation:** Stage 1 resampling on Adult data (F1 for class 1: 0.67 → 0.93); Stage 2 AIF360 adversarial debiasing (disparate impact 0 → 0.56).

Likely questions

1. Define bias in data and models; why is understanding bias important? (6)
2. Explain how bias can impact decision-making processes. (8)
3. Classify and explain the types of bias with examples. (8)
4. Describe any three real-world cases of biased AI models. (6)
5. Explain techniques to detect bias in AI systems. (8)
6. Explain techniques to mitigate bias; classify them into pre-, in- and post-processing. (10)
7. Describe the steps to implement bias detection and mitigation using resampling and AIF360 adversarial debiasing; interpret the metrics. (10)